

Which Reroute Will an Airline Accept? Learning Route Preferences from Revisions

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Abstract—When an airline revises the route in a flight plan, the revision records a choice it actually made: the route it dropped and the route it filed instead. Each such pair carries the key performance indicators of both routes (planned fuel, flight time, en-route charges and attributed delay) together with the waypoint sequence each route follows. This paper treats 1.48 million such pairs from eighteen months of European traffic as revealed preferences and trains a pairwise ranking model to predict which of two candidate routes an airline would choose. On a held-out test set the model never saw during training, it identifies the chosen route 68.1 % of the time, against 55.0 % for the strongest single-indicator rule. On the fifth of pairs where the model is most confident, it reaches 94.9 %. Because its confidence is calibrated, the model can support a proposal channel that offers fuel-saving routes only where they are likely to be accepted, at a precision chosen in advance. A simulation of such a channel against the airlines’ own later filings reaches 79.2 % precision and surfaces 20.1 kilotonnes of planned-fuel savings in a single quarter.

Index Terms—air traffic flow management, airline preferences, learning to rank, fuel efficiency

I. INTRODUCTION

On a spring morning in 2026, an Airbus A320 was filed from Amsterdam to Barcelona on a route carrying 102 minutes of attributed air traffic flow management (ATFM) delay, and the operator subsequently re-filed. The new route shed that delay completely, and it cost the airline on each of the three counts it pays for directly: 39 minutes of flight time, 1 397 kg of extra planned fuel, and 466 euros more in en-route charges. Both routes are held in the network’s records and so, implicitly, is the operator’s judgement that the delay would have cost it more than the detour did.

That judgement is the quantity this paper models. A flight plan is not filed once and left alone: an airline may revise it at any time until the aircraft leaves. An operator may reroute to avoid ATFM delay and prevent it from propagating to subsequent legs, or it may reroute because updated weather reveals a more fuel-efficient alternative. The archive records what changed rather than why, and every revision leaves the same trace: the route replaced and the route that replaced it.

The network records over a million such revisions a year across European airspace. A regulation may occasionally leave no viable alternative, but most revisions record a genuine preference: the operator had two routes available and filed the one it preferred. If those preferences could be predicted, a route proposal that saves fuel need not be offered blind; it could be aimed at the flights whose operators would file it.

Candidate routes are not hard to come by. What is hidden is how each operator weighs them: the balance between fuel, flight time, en-route charges and delay differs by airline and by day, and no airline publishes it. The terms of that trade-off, however, are recorded for every route in the network’s operational data, and so is the outcome: the route the airline actually filed. A preference model fitted to enough such observations can learn the underlying weighting without access to any operator’s cost function.

The contribution of this paper is threefold. First, it shows that airline route acceptance is learnable at network scale from revision events alone, and that the confidence of the resulting model is calibrated well enough to support a chosen operating point rather than a single accuracy figure. Secondly, it argues that the label construction is itself a design decision worth defending: a revision pair records a preference the airline actually expressed. The usual alternative ranks the filed route above candidates produced by a route generator, a comparison the operator may never have been shown. Thirdly, it reports what the model has learned rather than only how well it scores, namely that route structure outweighs any single cost indicator, together with the caveats that such an attribution demands.

The remainder of this manuscript is organised as follows. Section II explains why a pair of routes drawn from a revision constitutes a preference observation and what a model of such observations is for, and Section III positions the work within the literature on rerouting and route choice. Section IV describes the archive, the split protocol and the model, while Section V reports the results. Finally, Section VI concludes and outlines future work.

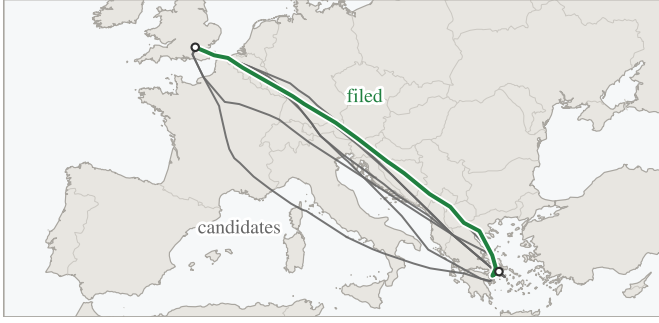
II. BACKGROUND

This section explains how a flight-plan revision becomes a labelled preference pair, why that label is preferred to one built from generated candidates, and what the model is for.

A. Flight-plan revisions as preference data

In the European network a flight plan is filed before departure and may be revised until the flight leaves. Every filing is a message, and so is every later change. The network keeps them all, so what it holds for a flight is the whole sequence of messages sent for it. A route change is accordingly a pair of consecutive messages rather than a record of its own: the message that changed the route, and the one it replaced.

(a) candidates



(b) a revision



Fig. 1. Two ways to build a training pair, on one city pair. Left: the filed route (solid green) is ranked above candidates a route generator would supply (grey); the grey tracks drawn here are real routes flown on the same city pair, standing in for generated ones. Right: a revision supplies two routes that the same operator filed for the same flight hours apart, one of which it abandoned. Only the right-hand pair records a preference the airline actually expressed.

The two together describe a decision, because the operator had committed to the earlier route and then filed the later one for the same flight, which makes the later route the one it chose and the earlier one the route it abandoned.

The sharpest of those decisions are made under flow management. When the traffic expected in a volume of airspace exceeds what can safely be handled there, whether because of demand, staffing, weather or equipment, a *regulation* is placed on it and the flights crossing it are given calculated take-off times at the rate it can absorb; the difference between a flight's calculated and requested take-off times is its ATFM delay.

An operator facing an ATFM delay has a choice that is genuinely economic: accept it, or file a route that avoids the constrained volume at the cost of distance, fuel and en-route charges, and neither answer is universally correct. For instance, a flight with a tight aircraft rotation may find forty minutes of delay far more expensive than a tonne of fuel, whereas the same delay on the last leg of the day may well be cheaper than the detour. This is why the decision is worth learning rather than deriving: the costs that determine it are operator-specific and flight-specific, and the network does not observe them, though the outcomes of those trades are recorded. Published European reference values [1] give the order of magnitude of the cost of delay and are widely used for analysis, but they are not the cost function any particular operator applies.

Most revisions, though, are not a response to flow management at all. Seven pairs in ten in this archive carry no regulation on either route (Table I), and within those the two routes differ by a median of one kilogramme of planned fuel.

The same events also settle how the training labels are built, because which of two routes an operator prefers cannot be asked at network scale but it can be observed, and the labels used herein follow the logic of revealed preference (i.e., inferring what an actor values from the choice it made when a concrete alternative was available) [2].

The alternative label construction, adopted by the authors' own earlier work, ranks the filed plan above candidates supplied by a route generator [3]. Its weakness is not that such routes are unrealistic. It is that nobody knows whether any of them ever reached the dispatcher who planned the flight.

The label asserts that the operator preferred the filed route to the generated one, whereas all the record shows is that the operator filed one route, so the comparison the label claims may never have taken place at all.

A revision pair claims nothing that did not happen, and Fig. 1 draws the two constructions side by side on one city pair. Both routes of a revision reached the dispatcher's screen, since the operator filed each of them in turn, for the same flight, on the same day. The route labelled less preferred is therefore not a plausible alternative constructed after the fact but one the operator had committed to and then abandoned, which makes the training signal a comparison that demonstrably took place.

Ranking a filed plan above generated candidates may require only that the model recognise which route looks like a filed plan, which is a different and much easier skill than knowing which route an operator would prefer. For instance, a generated candidate may join published segments in a way no dispatcher would ever file, so that telling it from a filed plan can be done on the shape of the route alone, without knowing anything about the operator that would fly it.

B. Two operational use cases

The first use case is a proposal channel: propose to an airline a route that is more fuel-efficient than the one it filed and that the model deems likely to be accepted, on the grounds that the operator may not have seen or evaluated that alternative in time. Calibrated confidence is what makes the channel viable: because the model's score estimates the true probability of acceptance, choosing a threshold directly sets the precision of the proposals. A stricter threshold means fewer proposals but a higher fraction that the airline would accept, so the channel can be tuned to a point where it stays credible rather than being ignored.

The second use case is overload release: when a traffic volume is overloaded and flights may be rerouted out of it, someone must decide which flights to approach first. At present, that ranking has no systematic basis in acceptance likelihood. Ranking flights by whether an acceptable and more fuel-efficient alternative exists would let the network manager relieve the overload while approaching fewer operators and asking less of each.

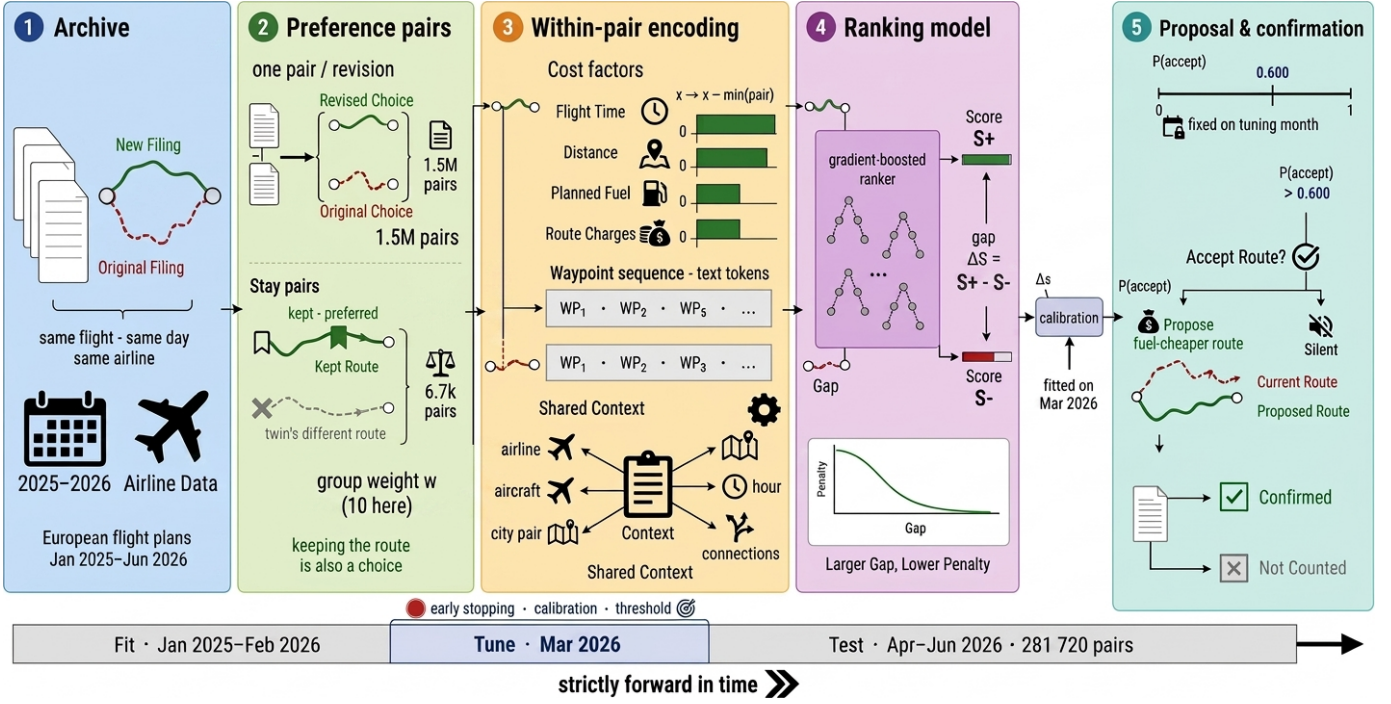


Fig. 2. How a filing archive becomes a proposal, above the fit, tune and test periods. The pair box shows both constructions: a revision, which is one flight and two routes, and a stay pair, which is two flights. The inset in the fourth stage is the training objective, whose penalty falls as the gap between the two scores widens and never reaches zero; the calibrated probability against its threshold is the only condition on a proposal. The bars of the third stage are one illustrative pair, in which the chosen route is the costlier on each indicator, the commoner direction on this archive rather than the invariable one.

Neither use case is tested end to end here: this paper demonstrates the preference model on which both depend, and simulates the first against the airlines' later filings.

III. RELATED WORK

Work on rerouting in ATFM has largely approached the problem from the network side. Over two decades ago, the authors of [4] formulated rerouting as a dynamic network flow problem, and the authors of [5] subsequently added a stochastic model that reroutes flights as capacity evolves. Both optimise an objective defined by the network and assume, in effect, that the resulting assignment is executed, so that the willingness of the airline to fly it is not modelled.

More recently, route choice in Europe was studied as a trade between fuel and en-route charges [6], which establishes that the choice is economic and therefore learnable, but uses no label in which the operator rejected an alternative it had itself filed. The closest work to the present one is that of the authors of [7], who incorporate airline preferences into collaborative rerouting decisions and show that accounting for them changes which reroutes are selected. Those preferences, however, are inputs to an optimisation, whereas this paper learns the preference function itself, at network scale, from what airlines did rather than from what they stated.

The direct predecessor of this work applies learning to rank to flight routes for a different purpose: predicting the flight plans that have not yet been filed, so that traffic demand can be estimated earlier than the filing horizon allows [3].

That work established the framing adopted herein, namely a gradient-boosted ranker that scores routes by their indicators in context, and it reported the very difficulty that motivates the present paper. Its training pairs take the filed plan as preferred and a set of generated proposals as less preferred, which is the construction that Section II argues against.

IV. METHODOLOGY

Figure 2 shows the procedure end to end; the two subsections that follow cover the data and the model.

A. Data and pair construction

Eighteen consecutive months of European network data, from January 2025 to June 2026, are used. After retaining only those revisions that change the horizontal route, and only those pairs with complete key performance indicators, the archive contains 1.48 million labelled pairs.

What those decisions look like matters before any model touches them, and Table I breaks the archive down by the regulation context in which each revision was made. The unregulated majority differs by a median of one kilogramme of planned fuel, and the pairs that cost fuel are elsewhere: the revisions that escape a delay outright, or reduce one, together account for 18.4 % of the archive.

A revision that escapes a delay moves to a route burning a median 101 kg more than the one it replaced, and that route burns less fuel in only 28.1 % of cases; a revision that merely reduces the delay costs a median 65 kg and is more fuel-efficient in 31.6 %.

TABLE I

THE REVISION ARCHIVE BY DECISION CONTEXT, OVER ALL EIGHTEEN MONTHS. A ROUTE COUNTS AS REGULATED WHEN THE FLIGHT, WHILE FILED ON THAT ROUTE, HAD A REGULATION ATTRIBUTED TO IT, WHICH LEAVES THE DELAY ITSELF FREE TO BE ZERO. UNREGULATED PAIRS CARRY NO REGULATION ON EITHER ROUTE, ESCAPES CARRY ONE ON THE ABANDONED ROUTE ONLY, THE NEXT TWO CARRY ONE ON BOTH, AND THE LAST ONLY ON THE NEWLY FILED ROUTE. THE FUEL DIFFERENCE IS BETWEEN THE TWO ROUTES OF A PAIR.

context	pairs	share [%]	median Δ fuel (new – old) [kg]	new route burns less [%]
unregulated	1 047 228	70.6	1	46.2
escaped delay	117 594	7.9	101	28.1
reduced delay	155 326	10.5	65	31.6
kept or raised delay	141 648	9.6	0	47.4
took on delay	21 337	1.4	–5	51.8

Each route is described by twenty-three features, summarised in Table II. The two routes of a pair are separated in time as well as in geography, however, and that must be addressed before anything is learned. The replacement is by construction the later message, so any feature that drifts with time identifies the answer without saying anything about the quality of the route. The time remaining to departure is the clearest case: it is necessarily smaller for the route that replaced the other, so a model given it in raw form could answer “the one filed later”, be right on essentially every pair, and know nothing whatever about routes. This is leakage rather than learning, and it would be invisible in the accuracy figure.

To ensure a fair comparison between the two routes, every flight-level attribute was therefore harmonised across the pair, both routes taking the value carried by the later message, so that no such attribute can separate them and only route-level quantities differ. The filing time is the case that matters most, and harmonising it in that way leaves both routes carrying the smaller of the two times to departure, which is what makes the shortcut disappear.

Two feature-construction decisions shaped the result more than the choice of learner did, and the first is that the four computed indicators enter as within-pair differences rather than magnitudes, obtained by subtracting the pair minimum from each, because absolute quantities are close to meaningless across the network: a 2 000 kg fuel figure means one thing on a short-haul leg and quite another on a long-haul sector, whereas the 300 kg by which one route exceeds the other carries the same meaning whether the sector is 400 or 4 000 km. Secondly, the ordered waypoint sequence is carried as text and encoded as a bag of tokens (i.e., each waypoint identifier is treated as a word and the route as the unordered collection of its words), which lets the model exploit route structure rather than aggregate distance alone. For instance, two routes on the same city pair may differ by under one per cent in distance and in planned fuel, and so be indistinguishable on every indicator, while only one of them crosses a volume whose identifier the model has met in thousands of earlier revisions; the token set carries that fact and the aggregate distance cannot. Section V-B shows that the first of the two matters far more than the authors expected.

TABLE II

WHAT THE MODEL SEES FOR EACH ROUTE; SOME ROWS SUMMARISE SEVERAL FEATURES. THE FIRST TWO GROUPS AND THE LAST BELONG TO THE ROUTE AND DIFFER WITHIN A PAIR, THE INDICATORS ENTERING AS DIFFERENCES FROM THE PAIR MINIMUM; THE ROTATION AND CONTEXT ROWS BELONG TO THE FLIGHT AND CARRY THE SAME VALUE ON BOTH ROUTES, THE ONE EXCEPTION BEING THE OUTBOUND CONNECTION TIME, WHICH IS SHORTENED ON THE SLOWER ROUTE BY THE DIFFERENCE IN FLIGHT TIME AND IS THEREFORE ROUTE-DEPENDENT AFTER ALL.

group	content
indicators	flight time, distance, planned fuel, route charges
regulation	attributed delay (own, inbound, outbound), their count, reason, and the type and identifier of the protected volume, meaning the airspace whose capacity the regulation protects
rotation	inbound and outbound connection times and states
context	operator, aircraft type, city pair, hour, weekday, month, time to departure
route	the ordered waypoint sequence, as text

Every revision pair is a flight that changed route. Most regulated revisions move away from delay (Table I), so the model sees delay far more often on the route the airline abandoned than on the one it filed instead, and learns to overweight it. That is a bias: flights that carried a delay and chose not to revise are absent from the archive altogether, so the model never learns that an airline can accept a regulation and keep its route.

A second, much smaller dataset was therefore constructed, in which *keeping* the route is the revealed choice. Let us consider two flights on the same day, with the same origin, destination and aircraft type, that flew different routes and neither of which revised its plan, and let us suppose that one of them was caught by a regulation while the other was not. The regulated flight had a reason to move and an alternative that was demonstrably flyable that day, namely the route its twin was flying, and yet it stayed, though the twin does not establish that the same routing was open at the stayed flight’s own departure hour. Its route is accordingly labelled as preferred, and that of its twin as the alternative it declined.

The yield is low: 6 672 such *stay pairs*, of which 1 093 fall in the test months, against 1.48 million revision pairs. Stay pairs are appended to the revision dataset for both training and evaluation, entering the fitting set at an increased group weight (the pair’s weight in the ranking loss). Section V-B compares the model trained with and without them.

The evaluation is strictly out of time, in the sense that the model is fitted on January 2025 through February 2026 (1 134 064 pairs), tuned on March 2026 (67 349), and evaluated once on April to June 2026 (281 720). Every threshold, calibrator and stopping decision is taken on the tuning month alone, and flights appearing in the tuning or evaluation months are removed from the fitting set.

All the data used is held by the network for operational purposes and was processed under existing data-handling arrangements. Operator identity enters the model as a categorical feature only, and no operator or airspace volume is named anywhere in this paper.

B. Model, training and calibration

The operational question is comparative: given two routes for one flight, which of them does the airline prefer? Accordingly, the model gives each route a single number, which is not a probability and has no units, and which is only meaningful next to the number given to another route for the same flight. Training pushes those numbers apart in the right direction. For every historical pair the model pays a penalty, largest when the abandoned route outscores the chosen one; the penalty falls as the chosen route's lead grows and never quite reaches zero, so the model keeps pressing even once the order is right [8]. Nothing in the objective asks the model to reproduce a route in isolation, which is what makes the pair the natural unit.

Gradient-boosted decision trees are used as the scoring function [9], at a depth of 8, with the learning rate selected automatically and with early stopping on the tuning month. Stay pairs enter the fitting set with a group weight w , swept over $\{10, 30, 100\}$; Section V-B reports why 10 is used.

Once trained, the model has still to be connected to a decision. Let us consider a real pair from the held-out quarter, on which the model scores the two routes -2.82 and $+0.90$. Neither number means anything on its own, since the score has no units, and the same two numbers on another flight would carry no comparable meaning. What can be read is their difference, here 3.7 in favour of the second route. Even that is not yet a decision, because nobody in an operations room knows what a gap of 3.7 is worth.

Calibration is the step that turns the gap into a statement of the form “about eight times in ten, the route the model prefers is the one the airline goes on to file”. It is a monotone mapping from score gap to the observed rate at which the preferred route was the one filed, fitted on the tuning month alone [10] and applied unchanged to the evaluation months. Isotonic regression was selected there on log loss, against the raw sigmoid of the score gap and a two-parameter Platt fit, before those months were touched.

Models of this kind usually over- or under-state their confidence and benefit from such a fit [11]. This one did not. On the held-out quarter its own raw sigmoid was better calibrated than the isotonic fit chosen to correct it: the sigmoid missed the observed rate by 0.9 percentage points on average, the isotonic fit by 1.8. The choice was fixed before those months were opened and was not revised afterwards, since revising it on held-out evidence is exactly the peeking the protocol forbids, so on this archive the fitted mapping is a safeguard rather than a correction. What matters operationally is that stated and observed rates agree to within those 1.8 points, because without calibration one can rank proposals but cannot decide how many of them to make.

The policy that results is short. A flight is eligible whenever its two routes differ in planned fuel at all, which holds for 97.9 % of the held-out pairs, and the one burning less fuel is the candidate. That candidate is proposed when its calibrated acceptance probability exceeds a threshold τ , selected on the

tuning month for a target precision and reported unchanged on the evaluation months; $\tau = 0.600$ throughout what follows.

The calibrator always scores the route the model prefers, so the probability it returns is by construction at least a half. Where the more fuel-efficient route is the one the model ranks lower, the channel must therefore use one minus that probability, which is below a half, so such a pair is never proposed at any threshold of a half or above. A proposal is realised when the airline's own filing chose that route.

V. RESULTS

This section reports how accurately the model predicts airline choices, whether the change bias of Section IV is controlled, what the model has learned about route preference, and what a proposal channel built on it would deliver.

A. Predictive performance

On the held-out quarter (April to June 2026, 281 720 pairs) the model identifies the airline's chosen route 68.1 % of the time (95 % interval 67.9–68.3), against 50 % for random choice and 55.0 % for the best single-indicator rule.

No single indicator captures much of the decision. A rule preferring the route with less attributed delay reaches 71.6 % on the pairs where both routes carry a delay figure, but those are only 14.6 % of the archive, so pooled over all pairs it scores 53.2 %. The task is irreducibly multi-dimensional: airlines weigh delay, flight time, fuel and charges against one another, and no one quantity dominates.

Not all of those predictions carry the same weight. The model's confidence in each case is the gap between the scores it assigns to the two routes: a wide gap means a confident prediction, a narrow gap means the pair is close to a coin toss. As Section IV-B describes, a calibrator converts that gap into a probability, so that the model can say not only which route the airline would choose but how likely it is to be right. On the fifth of pairs where that confidence is highest, the model is right 94.9 % of the time. Fig. 3 traces the full trade-off: as the model abstains on low-confidence cases, the share of cases it answers falls but its accuracy rises. A channel need not answer every flight, so this curve, rather than the pooled 68.1 %, is what characterises the system.

Where those 68.1 % sit is more informative than the pooled figure, and Table III splits the quarter three ways. Accuracy climbs steeply with the delay carried by the abandoned route, from 63.9 % where there was none to 84.5 % between half an hour and an hour, where the climb stops. That is what an economic reading predicts, since the larger the delay, the clearer the incentive and the easier the decision is to predict.

A model that scores well on average but decays from month to month is not deployable, and across April, May and June accuracy moves between 67.5 % and 69.0 %. A quarter of evidence rules out rapid decay, but it does not establish the retraining cadence a deployed system would need.

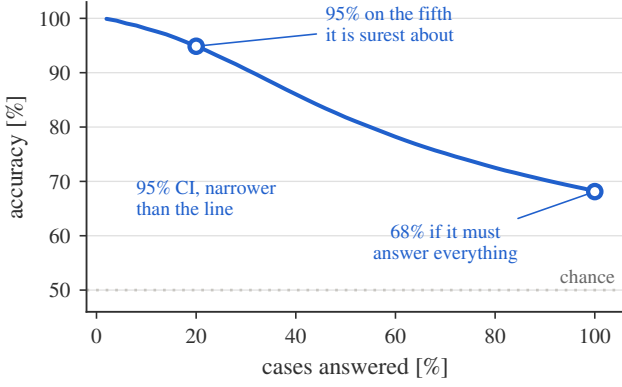


Fig. 3. Accuracy against the share of cases answered, on the held-out quarter, a case being one pair of routes. Moving left, the model answers fewer cases, lowest confidence dropped first, and is right more often; the dotted line is chance.

TABLE III

ACCURACY BY STRATUM ON THE HELD-OUT QUARTER, WITH WILSON 95 % INTERVALS. THE TWO BLOCKS SPLIT THE SAME 281 720 PAIRS BY REGULATION EXPOSURE AND BY THE DELAY ATTRIBUTED TO THE ABANDONED ROUTE.

stratum	pairs	accuracy [%]
<i>which route carried a regulation</i>		
neither	178 370	64.0 [63.8, 64.2]
old route only	30 953	88.0 [87.6, 88.3]
both	67 587	70.6 [70.2, 70.9]
new route only	4 810	58.6 [57.2, 60.0]
<i>delay on the abandoned route</i>		
none	205 205	63.9 [63.7, 64.1]
up to 15 min	19 672	70.3 [69.7, 71.0]
15 to 30 min	18 283	79.3 [78.7, 79.9]
30 to 60 min	21 231	84.5 [84.0, 85.0]
above 60 min	17 329	83.5 [82.9, 84.0]

B. Change bias

Section IV noted that most regulated revisions in the archive move away from delay, so a model trained only on revisions overweights delay as a negative signal and never sees the flights that accepted a delay and kept their route. Stay pairs correct this by adding such flights to both the training and the evaluation set.

The effect is clear on both populations. Trained on revisions alone, the model scores 68.9 % on the revision pairs of the held-out quarter but only 34.8 % on the 1 093 stay pairs: whenever a flight stayed despite a delay, the model predicts the wrong route. Adding stay pairs at group weight 10, meaning each stay pair counts ten times as heavily as one revision pair in the ranking loss, raises the stay-pair accuracy to 90.0 % and brings the revision accuracy to 68.1 %, a cost of 0.8 points. Table III confirms that the adopted model also scores 88.0 % on flights that changed route after a regulation (the *old route only* stratum), so the correction does not trade one population for the other.

C. Error analysis and feature attribution

Errors concentrate where the model is already unsure. Among the quarter of pairs whose two routes score furthest

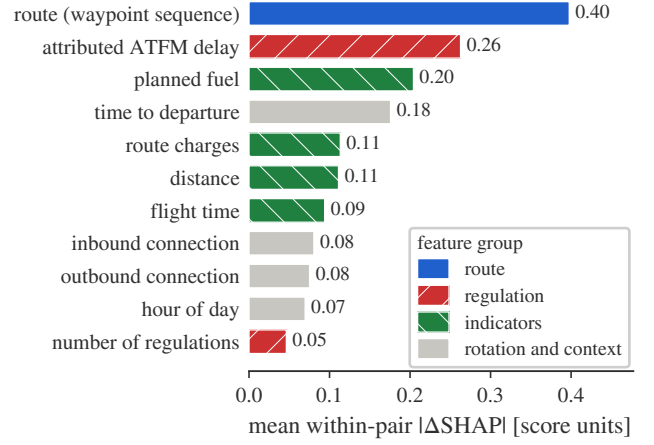


Fig. 4. What moves a decision: the mean absolute *difference* between a feature's Shapley value on the two routes of a pair, over the held-out quarter, with the rotation and context groups of Table II drawn together. The waypoint sequence is the largest single term, ahead of every *individual* cost indicator, though the four taken together sum to more; attributed ATFM delay outweighs planned fuel. Rotation and context features are identical on both routes, so their bars measure interaction rather than preference.

apart, only 5.6 % of all errors fall, against the 25 % expected if error were independent of confidence. In practical terms, most wrong predictions are cases the model was nearly indifferent about, and a channel that abstains on those cases avoids most of its mistakes. The expensive errors are the opposite kind: the model confidently proposes a route that the operator declines. A deployed channel has no direct measure of that rate, because confident errors on stay pairs, where the model wrongly advises leaving a route the airline kept, are the closest available proxy, and that rate is the 10.0 % complement of the 90.0 % of Section V-B.

A ranking model an operational user is asked to trust must be open to interrogation, and Shapley attributions were therefore computed over the held-out quarter [12]. Because the model decides on a comparison rather than on one route, what Fig. 4 reports is the *difference* between a feature's attribution on the two routes of a pair, averaged in magnitude over the quarter: both routes receive Shapley values for each feature, and the plotted quantity is how much more the feature contributed to the preferred route's score than to the other's. That is a ranking not of what correlates with acceptance, but of what moves this particular model's decisions.

On that scale the waypoint sequence moves the score by 0.40, attributed ATFM delay by 0.26 and planned fuel by 0.20, so route structure carries information about the choice that the cost indicators do not.

The waypoint text ranks first by attribution, yet removing it and retraining costs only 1.4 points of accuracy (68.2 against 66.8). The two numbers answer different questions: attribution measures how much the feature moves the score of the model that has it, while ablation measures how much worse a model trained without it would be. Where features are correlated, as route geometry and cost indicators plainly are, the model trained without the waypoint text recovers part of the lost

information through the remaining features, so the ablation cost is smaller than the attribution rank would suggest.

D. What a channel would deliver, and the conditions for it

The question that remains is what a proposal channel built on this model would deliver in practice. Three conditions stand between these results and deployment. First, the delay attributed to a flight depends on which regulations are active at the time the model scores it, and that picture changes continuously as volumes open, close or shift their rates, so the channel must score with the regulation state of the moment rather than with a snapshot that may already be stale; the constraint is data freshness, not computation. Secondly, a deployed channel would see every flight, including the majority that never revise their plans, a population on which the model was evaluated only through the 6672 stay pairs of Section V-B. Thirdly, acceptance itself is never observed in this study; every figure rests on what the airline filed next rather than on a response to an actual proposal.

At the operating point fixed in advance for 80 % precision ($\tau = 0.600$, delivering 79.2 %), the simulated channel surfaces 56 162 proposals over the held-out quarter. Those proposals carry 20.1 kt of planned-fuel savings, counted only where the airline’s own later filing chose the route proposed to it, or 63.5 kt of CO₂ at 3.16 kg per kilogramme of fuel. Annualised, that is between 222 and 254 kt of CO₂. The two bounds differ because the held-out quarter falls in spring and summer, when traffic is above the annual average: multiplying the quarter by four gives 254 kt, while applying the per-flight saving rate to twelve months of eligible traffic gives 222 kt. The channel figures are computed on flights that revised their plans, and planned fuel is the dispatcher’s estimate filed before departure, not the fuel actually burned, so they should be read as what a channel could have surfaced earlier rather than as a traffic-wide counterfactual. The full channel analysis is the subject of a journal extension of this work.

Figure 5 follows one of these proposals through the scoring pipeline. A narrow-body was filed on a medium-haul route with no regulation; the alternative saves 2016 kg of planned fuel at essentially the same flight time but 227 EUR more in route charges. The model scores the alternative 3.7 above the filed route, the calibrator maps that gap to a 99 % acceptance probability, well above $\tau = 0.600$, and the channel proposes. The airline’s later filing chose exactly that route. The margin is far from typical: the median confirmed proposal saves 130 kg, but this case shows every step of the pipeline in one place.

VI. CONCLUSIONS

Airline route acceptance is learnable from flight-plan revisions at network scale. A model fitted on 1 134 000 revisions identifies the chosen route in 68.1 % of held-out cases, 13.1 points above the best rule built on any one indicator, and its confidence is informative enough to choose where to operate rather than settle for a single accuracy figure: on the fifth of decisions it is most confident about, it is right 94.9 % of the time. The model weighs route structure more heavily than

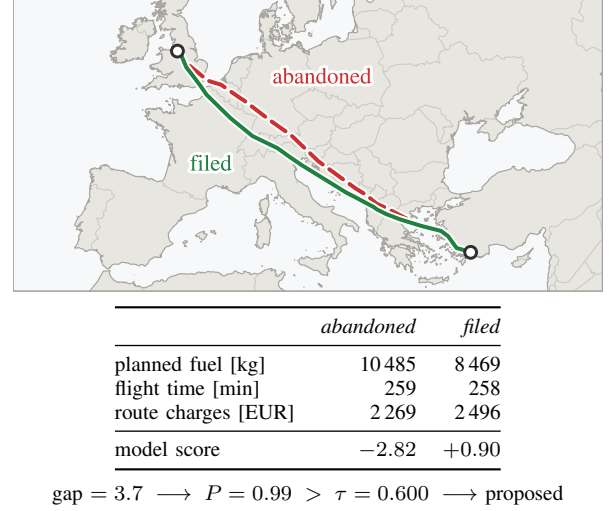


Fig. 5. A worked example from the held-out quarter. The route initially on file (dashed red) and the fuel-saving alternative (solid green) are shown on an anonymised city pair. The alternative saves 2016 kg of planned fuel at 227 EUR more in route charges and the same flight time. The calibrated acceptance probability exceeds the threshold; the airline’s own later filing chose the proposed route.

any individual cost indicator, though removing the waypoint text costs only 1.4 points, so what it contributes is real but secondary. It is nonetheless information that a formulation based on costs alone cannot represent.

The number that characterises this system is not 68.1 % but the curve of Fig. 3, because a channel chooses where on that curve to sit, and because ranking a pair correctly and proposing correctly are different quantities of which the second is the operational one: at the threshold used here, the channel proposes on about a fifth of the flights whose two routes differ in planned fuel at all, and 79.2 % of those proposals match what the airline went on to file. Accordingly, a first deployment would sit at low coverage on Fig. 3, answering only the cases it is most confident about, and would widen that coverage only as acceptance is observed.

Two bounds remain. The first is the population the second deployment condition names: the data on flights that stayed is small, 6672 pairs in all and 1093 in the test months, and each stay pair joins two different flights rather than two routes filed for the same flight, so a model could learn to recognise how the pair was built rather than the preference it encodes. The stay-pair accuracy of 90.0 % in Section V-B shows that the model is not simply reading regulation status, but it cannot rule out subtler artefacts of the construction.

The second lies in how a route is represented. The model reads a route as the ordered sequence of waypoint names along it, so what it holds about route structure is a vocabulary, not a geometry. Those names are European: a waypoint such as KONAN or MAROT exists only in this region’s airspace, so little of the 1.4 points the waypoint text is worth would survive a move to another continent. They are also revised periodically, so a piece of airspace can return under a name the model has never seen.

We tested the obvious alternative: a Transformer encoder was trained on geometric route descriptors (shape alone, no names) and its embeddings were fed as additional features into CatBoost. The hybrid scored below the model that uses the text tokens directly, and a linear classifier on the embedding differences separated preferred from abandoned routes at little better than chance, which points at the geometric input rather than the training objective as the limitation. A name-independent representation of route shape is nonetheless what would make this model portable, and remains worth pursuing.

A related limitation is that fuel appears in the feature set as kilograms of planned fuel, not as a cost. A kilogram of jet fuel in 2025 and a kilogram in 2035 will not carry the same price, so the relative weight an airline places on fuel against, say, route charges in euros will shift as fuel costs change. Recasting fuel as a monetary cost, or normalising all indicators to a common unit, would make the model's trade-offs more stable across years.

Future work should therefore begin with a balanced stay dataset in which regulation status is independent of the label, which requires data that is not currently assembled rather than data that does not exist. A shadow-mode trial logging the real responses of operators would close the remaining gap.

DISCLAIMER

The views expressed are those of the authors and do not necessarily reflect the official position of EUROCONTROL or of its Member States. This paper reports research and does not commit to any operational service.

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